The Pareto Landscape of Moral Reasoning

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Abstract

Evaluating moral reasoning is hard because the moral domain lacks the ground truth that organises evaluation elsewhere: competent reasoners can start from similar intuitions and reach different defensible conclusions. We argue that this is not a measurement failure but a structural feature, and we propose a conceptual framework that takes it seriously. Moral reasoning, on our account, aims at a *good epistemic state*: a pair of commitments and principles that hangs together. Such a state is graded along two axis families that genuinely conflict: epistemic desiderata (account, systematicity, faithfulness) and value loadings (freedom, well-being, justice, dignity, solidarity). The rational object of study is then the *Pareto frontier* of attainable epistemic states. We develop a vocabulary for the shapes this frontier takes (convex segments, concave segments, knees, plateaus, corner attractors, dominated regions), and we recast familiar reasoning methods — reflective equilibrium foremost among them — as strategies for navigating the landscape toward the frontier. We sketch the corpus that would let this framework be studied empirically, and outline what such study would reveal about moral reasoning in humans and in language models.

1 Motivation

Current alignment work focuses on the moral *content* a model holds: constitutional principles, character training, value specifications. Less attention has gone to moral reasoning as a *capacity* (Carlsmith, 2026). But a value-aligned model whose reasoning fails when its values come into tension is still unsafe: it can produce outputs that look like principled moral thought without being so. Diagnosing such failures is hard because the moral domain lacks a ground truth against which to score answers (Queloz, 2025): competent reasoners can reach different conclusions without either being simply wrong.

This paper proposes a framework that takes the absence of ground truth as a *structural feature* rather than a measurement failure. We articulate what moral reasoning aims at, what makes its outputs better or worse without a single right answer, and how the multi-objective optimisation literature gives us a precise vocabulary for the kinds of trade-offs the moral domain actually exhibits. Familiar methods such as reflective equilibrium then reappear as *strategies for navigating* this landscape, comparable in principle to alternative strategies and amenable to empirical study once a suitable corpus is in place.

2 Conceptualising moral reasoning

An **ethical situation** confronts an agent with a decision: several actions are open, and none is clearly right or wrong. Moral reasoning is what the agent does between encountering the situation and acting. We decompose it into identifiable stages:

1. *Identify ethically relevant properties* of the situation — who is affected, which interests are at stake, the nature of the action on offer.

Preprint. Under review.

2. *Form initial commitments* about how to act, on the basis of those properties. These are the agent's pre-theoretical verdicts about the case, the starting material of further reasoning.
3. *Propose principles*: generalise across one's commitments and find rules that would justify them. This is the inductive or abductive move.
4. *Deduce*: from a candidate principle, derive the commitments it would imply across other situations.
5. *Revise* when principle and commitment conflict, in either direction — the dialectical move at the heart of reflective equilibrium (Rawls, 1971; Beisbart et al., 2021).
6. *Evaluate* the resulting pair of commitments and principles.

The output of moral reasoning is an **epistemic state** $\sigma = (C, P)$: a set of commitments and a set of principles that hang together. We take the goal of moral reasoning to be reaching a *good* epistemic state. The question is what *good* means in a domain without a single right answer.

3 What makes an epistemic state good

Two axis families grade σ simultaneously. They genuinely conflict, in the sense that improving one typically costs another.

Epistemic axes. Properties of σ as reasoning, drawn from the formal account of Beisbart et al. (2021).

- ACCOUNT: how much the principles in P entail or support the commitments in C, penalising contradictions, gaps, and surplus content.
- SYSTEMATICITY: how simple, unifying, and broadly applicable P is, as opposed to a hodgepodge of case-by-case stipulations.
- FAITHFULNESS: how well the current commitments respect the agent's initial intuitions C_0, penalising arbitrary drift.

Ethical axes. Properties of σ as a moral stance. Each principle $p \in P$ carries a fuzzy *value loading* $\boldsymbol{v}(p) \in [0,1]^K$ over a small set of basic moral values — we work with five: FREEDOM, WELL-BEING, JUSTICE, DIGNITY, and SOLIDARITY, chosen so that the central tensions of moral life (liberty vs. community, fairness vs. welfare, welfare vs. dignity) appear directly as trade-offs among axes. The state σ inherits an aggregate value position $V(\sigma) \in [0,1]^K$.

Two families, two axes of disagreement. The epistemic family is broadly agent-independent: every reasoner has some stake in coherence. The ethical family is agent-dependent: different agents weight different values, and this pluralism is a constitutive feature of the moral domain rather than noise to be averaged out. A *good* epistemic state must do well on both, but "well" on the ethical family is relative to a value profile.

4 The space in which moral reasoning evolves

We can now state the framework precisely. Let Σ be the set of all internally consistent epistemic states. Each $\sigma \in \Sigma$ has a position in the joint objective space

$$\big(A(\sigma),\, S(\sigma),\, F(\sigma \mid C_0),\, V_1(\sigma), \ldots, V_K(\sigma)\big) \in [0,1]^{3+K}.$$

Multi-objective optimisation (Deb, 2001) gives us the right vocabulary for the structure of this space when its axes genuinely conflict.

Pareto dominance. A state σ' *Pareto dominates* σ, written $\sigma \prec \sigma'$, if every objective at σ' is at least as large as at σ and at least one is strictly larger.

The Pareto frontier. The frontier Π is the set of non-dominated states in Σ:

$$\Pi \;=\; \{\, \sigma \in \Sigma : \nexists\, \sigma' \in \Sigma \text{ with } \sigma \prec \sigma' \,\}.$$

Π is what we mean by *the space of defensible epistemic states*. For any state off the frontier, some defensible move strictly improves it on every axis the agent cares about; for any state on the frontier, no such move exists.

Reasoning traces. A reasoning trace is a finite sequence $\sigma_0, \sigma_1, \ldots, \sigma_n$ of epistemic states linked by moves of the kinds enumerated in Section 2. The trace has a *shape* in the joint space, and each move has a characteristic contribution: principle proposal typically lifts ACCOUNT and SYSTEMATICITY together; commitment revision trades FAITHFULNESS for ACCOUNT; principle revision trades SYSTEMATICITY for ACCOUNT, or one value for another.

Vocabulary for the shape of the frontier. The multi-objective literature has names for the features the moral landscape exhibits.

- **Convex segments.** Weighted sums of axes pick a unique answer. There is little genuine disagreement: reasoners differ in weights but agree in conclusion.
- **Concave segments.** Incomparable defensible answers: no weighting of axes selects between them. *This is where genuine pluralism lives.*
- **Knee points.** A small concession on one axis buys a large gain on another. The *natural compromise* of the case — where reasoning tends to land when neither value dominates.
- **Plateaus.** Many near-equivalent positions. The choice is under-determined and competent reasoners may land in slightly different places without error.
- **Corner attractors.** Extreme positions where one axis dominates and others collapse. Pure utility, pure autonomy, pure procedural justice. Often degenerate.
- **Dominated region.** Bad reasoning. Off-frontier states are off-frontier *because* some defensible move strictly improves them.

A worked landscape: medication adherence. A user is unwell and refuses a prescribed medication. The refusal is partly driven by delusional thinking. An LLM acting as a therapeutic interlocutor must decide how to respond. The ethically relevant properties include the user's autonomy, the medical risk, the cognitive basis of the refusal, and the asymmetry between agent and user. The initial commitments cluster around respecting autonomy, protecting well-being, not manipulating, and not abandoning. The values primarily in tension are FREEDOM and WELL-BEING.

Two defensible responses sit on a concave segment of Π. An autonomy-leaning response states the medical facts plainly, documents concerns, and does not push further. A care-leaning response explores the refusal through dialogue, attempts non-manipulative persuasion grounded in the user's stated values, and escalates to a clinician if risk warrants. Neither dominates the other under any weighting of FREEDOM and WELL-BEING; the segment between them is the locus of genuine pluralism.

An interior knee point may exist. A specific autonomy-preserving care strategy — naming the delusion gently, inviting the user's own values into the deliberation — might buy a large WELL-BEING gain for a small FREEDOM concession. Where the case admits such a knee, it is a natural compromise.

Several responses are off-frontier. Insulting or shaming the user; manipulating without succeeding (high FREEDOM cost, no WELL-BEING gain); lying about unrelated medical facts (low DIGNITY, low SYSTEMATICITY); walking away with no documentation or escalation (low ACCOUNT relative to the agent's stated commitments). Each of these is strictly dominated by a nearby frontier state, and that domination is what makes it bad reasoning.

5 Strategies for navigating the space

If moral reasoning aims at the frontier, the natural follow-up question is *how an agent gets there*. Reflective equilibrium is the philosopher-favoured answer, but it is one strategy among several. We sketch the space of strategies and the primitives they share.

Shared primitives. Any strategy that walks toward Π requires three capacities, independent of how those capacities are combined: *principle proposal* (induction from commitments to a candidate principle), *principle deduction* (derivation of commitments implied by a principle), and *coherence checking* (assessment of σ on the epistemic and ethical axes). For an artificial moral reasoner, these are the capacities to evaluate.

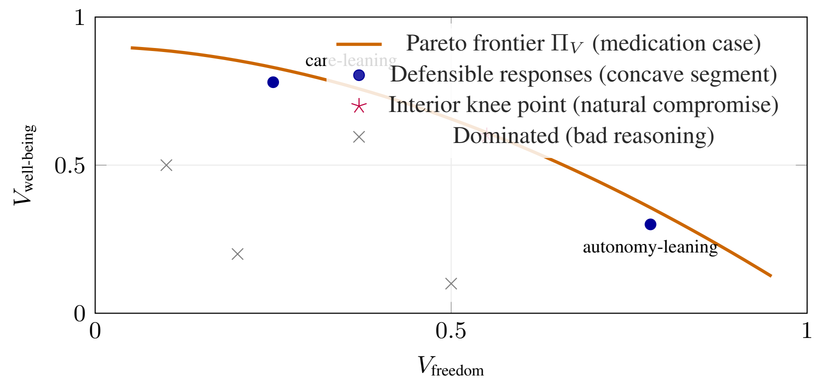

Figure 1: Sketch of the ethical-value Pareto frontier in the medication case, projected onto FREEDOM and WELL-BEING. Two defensibly different responses sit on a concave segment; an interior knee point captures a natural compromise; several reachable responses are strictly dominated.

Reflective equilibrium. Beginning from the initial commitments C_0 and an arbitrary T, alternate revisions of T and C, each step guided by current scores (Beisbart et al., 2021). The trace is local: each move depends on the current state, and the trajectory walks toward a fixed point. Reflective equilibrium is well-suited to finite reasoners with strong intuitions and limited search budget; it is the canonical strategy in the philosophical literature.

Sample-and-prune. Generate many candidate epistemic states by combining principles and commitments, score each, retain the non-dominated ones. A direct attempt to map a region of Π rather than walk toward a single point. Well-suited to reasoners with cheap sampling, perhaps less well-suited to finite human deliberation.

Tree search. Build a tree of reasoning moves from σ_0, expand promising branches, backtrack from dead ends. A natural fit for artificial reasoners with a notion of branch quality and a backtracking mechanism.

Single-pass abduction. Propose one principle from C_0, deduce its consequences, accept the result. The naive baseline. Often under-evaluated against the others.

Multi-start equilibrium. Run reflective equilibrium from several distinct starting points to discover multiple regions of Π at once. Particularly useful when the frontier is expected to be concave or plateau-shaped.

Corner-collapse strategies. Maximise a single axis (pure ACCOUNT, pure WELFARE, pure AUTONOMY). Degenerate in general but instructive as ablations and as failure modes of stronger strategies.

Which strategy for which reasoner? For humans, reflective equilibrium is the canonical answer, in part because human deliberation has a budget that favours local moves over wide sampling. For artificial reasoners — LLMs in particular — the question is open. Sampling-based strategies may exploit parallelism that humans cannot; tree-search strategies may map richer landscapes; the failure modes of single-pass strategies may or may not appear under chain-of-thought. The framework lets us ask these questions on equal footing once we can locate any strategy's output in the space.

6 A corpus to make this empirical

The framework so far is conceptual; turning it empirical requires a corpus. We sketch what we would dream of having and what it would let us do.

Contents. For a body of ethical situations, the corpus would contain three layers.

1. **Commitment pool** $\mathcal{C}$: candidate verdicts on each situation. Per-case action judgments of the form "in situation s, action a is required / permissible / forbidden," generated broadly enough to cover the defensible verdicts plus a margin of bad ones.
2. **Principle pool** $\mathcal{P}$: candidate moral principles, each tagged with a *fuzzy value loading* $\boldsymbol{v}(p) \in [0, 1]^K$ indicating to what degree the principle expresses each value. Optionally, a secondary tagging by ethical framework $\boldsymbol{f}(p) \in [0, 1]^M$ (utilitarian, Kantian, virtue, care, contractualist) supports framework-level analyses without committing to frameworks as primary axes.
3. **Dialectical graph** ρ: for every principle-commitment pair, a *fuzzy inferential strength* $\rho(p, c) \in [-1, 1]$, where positive values indicate degree of entailment, negative values degree of contradiction, and zero silence. This generalises a discrete entails / contradicts / silent labelling and captures the partial support relations that are typical in moral reasoning.

Construction. The corpus would be generated by a large language model and verified by philosophers at every stage — generation, fusion of near-duplicates, deletion of weak candidates, scoring of independent quality dimensions, value tagging, and inferential strengths. We treat the corpus as a reusable resource in its own right, independent of any downstream reasoning method that operates on it.

What it would let us do. Given a starting set of intuitions C_0 drawn from $\mathcal{C}$, the corpus lets us:

- Compute the Pareto fronts Π_E and Π_V of attainable epistemic states, and characterise their shapes locally and globally.
- Locate any epistemic state σ in the joint space, and say whether it is on Π, near Π, or dominated — and if dominated, by what.
- Inspect reasoning traces move by move: is each step principled (a justified abduction, a defensible commitment revision) or unprincipled (a faithfulness violation with no compensating gain)? Does the trajectory walk toward Π or away from it?
- Compare reasoning strategies on equal footing.

7 What this lets us study

The framework, combined with such a corpus, opens several lines of empirical work that are difficult to pose otherwise.

The geometry of moral disagreement. For a given case, is the local shape of Π convex (resolvable disagreement) or concave (genuine pluralism)? Does the case admit a knee point (a natural compromise)? Across many cases, what shapes recur, and what does that tell us about the structure of the moral domain?

Strategy comparison. Do different reasoning strategies reach different regions of Π on the same case? Does reflective equilibrium land in plateaus while sample-and-prune maps concave segments? Which strategy is more robust under reframing of the case?

LLM moral reasoning. A value-aligned model whose traces sit off Π in characteristic ways is a value-aligned model that reasons badly — and the framework lets us name how. Specific failure modes become diagnosable: *corner collapse* (the trace ends at an axis extreme), *anti-revision rigidity* (no move revises the principle side), *faithfulness violations* (commitments dropped with no justifying gain), *stability collapse* (different landings under semantically equivalent rephrasings of the situation). These are empirically testable hypotheses about LLM behaviour, but they only become testable once the framework names them.

The capacities that matter. If the shared primitives of Section 5 are the irreducible capacities of moral reasoning, then the question for any artificial moral reasoner is: which of those capacities does it have, and at what quality? The corpus lets us isolate each primitive — principle proposal, principle deduction, coherence checking — and evaluate it independently of the strategy it is embedded in.

8 Discussion

We have argued that moral reasoning is best framed not as the search for a single right answer but as navigation of a multi-objective landscape whose Pareto frontier is the space of defensible answers. The framework gives familiar pieces a clearer place: the goal of moral reasoning is a good epistemic state; goodness is graded along epistemic and ethical axes that genuinely conflict; the rational object of study is the Pareto frontier of attainable states; reasoning methods such as reflective equilibrium are strategies for navigating this landscape, alongside other strategies that may suit artificial reasoners better. The descriptive vocabulary of multi-objective optimisation lets us name what we see: convex and concave segments, knee points, plateaus, corner attractors, dominated regions — and the failure modes of reasoners that get stuck in any of them.

The work the framework still requires is the corpus (Section 6) and the empirical study it would support (Section 7). Neither is small, but both are now well-posed: the framework tells us what to build and what to measure.

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